Week 7 Assignment

T-tests and ANOVA

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf`, along with your Data Safari submission, via Canvas before Week 9′s lecture (Monday, following Spring Break). See the syllabus for the late policy.

Note: This week also includes a Data Safari exercise — a separate, short assignment. Both are due at the same time.

Part 1: Conceptual

1. You have two groups and want to compare their means. What determines whether you should use a paired or unpaired t-test?
2. You have four groups. Why is running six separate pairwise t-tests a worse idea than a single one-way ANOVA followed by post-hoc comparisons?

Part 2: Applied

Using the built-in `PlantGrowth` dataset:

1. Fit a one-way ANOVA of `weight` by `group`.
2. Check the assumptions (residual diagnostic plots, Levene's test).
3. Run a post-hoc comparison with Tukey correction.
4. Write a 2–3 sentence results-section-style summary of your findings (report the F-statistic, degrees of freedom, p-value, and which pairwise comparisons — if any — are significant).